EXPERIMENT NO: 01.

20/7/26

NETWORK COMMANDS USED IN LINUX & WINDOWS

Aim: To study various network commands used in Linux & windows.

Basic Networking Commands:

1) arp-a : short of address resolution protocol, it will show IP address of computer along with IP address & MAC address of router.

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DESKTOP-L7R3IN3.mshome.net (172.17.240.1) at 00:15:5d:7b:18:f5 [ether] on eth0

2) hostname: simplest of all TCP/IP commands. It simply displays name of your computer

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DESKTOP-L7R3IN3

3) ifconfig : displays detailed configuration information about your TCP/IP connection

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eth0: flags=4163<UP, BROADCAST, RUNNING, MULTICAST> mtu 1500
inet 172.17.249.113 netmask 255.255.240.0 broadcast 172.17.255.255

lo: flags=73<UP, LOOPBACK, RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10<host>
loop txqueuelen 1000 (Local Loopback)

4) netstat : (network statistics) displays a variety of statistics about a computers active TCP/IP connections.

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Activ internet connections (w/o servers)

Proto	Recv-Q	Send-Q	Local Address	Foreign Address	State

Active UNIX domain sockets (w/o servers)

Proto	RefCnt	Flags	Type	State	I-Node	Path
unix	3	[]	DGRAM	CONNECTED	1302	
unix	3	[]	DGRAM	CONNECTED	6322	
unix	2	[]	DGRAM	CONNECTED	3685	

5) nbtstat -a : solve problems with NetBIOS name resolution

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nbtstat -a DESKTOP-L7R3IN3

\Device\NetBT_Tcpip_{C3D19B6D-4DA1-4C26-8692-3CF211AE9211}

Node IpAddress: [0.0.0.0] Scope Id: []

Host not found.

VMware Network Adapter VMnet1:

Node IpAddress: [192.168.106.1] Scope Id: []

Host not found

6) ipconfig : displays detailed configuration details

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Windows IP configuration

Ethernet adapter Ethernet 2:

Connection-specific DNS suffix . :
Link-local IPv6 Address . . . . : fe80::1406:332:
d3c0:6c17.18

7) nslookup: (name server lookup) is a tool used to perform DNS lookups in Linux. used to display DNS details.

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nslookup www.google.com
Server: 172.16.96.1
Address: 172.16.96.1#53

Non-authoritative answer:
Name: www.google.com
Address: 142.251.156.119
~~Name~~

8) pathping: unique to windows, & is basically a combination of the ping & Tracert commands.

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pathping google.com
Tracing route to google.com [142.250.67.46]
over a maximum of 30 hops:

0 pikibot [172.16.102.158]
1 172.16.96.1
2 137.97.163.129
3 * * *

Computing statistics for 50 seconds...

a) **Ping**: (Packet INternet Groper) command is the best way to test connectivity between 2 nodes.

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Pinging google.com [142.250.67.46] with 32 bytes of data:

Reply from 142.250.67.46: bytes=32 time=17ms TTL=118

Ping statistics for 142.250.67.46:

Packets: Sent=4, Received=4, Lost=0 (0% loss).

10) **Route**: route command is used to show/manipulate the IP routing table.

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route print

Interface List

18...0a 00 27 00 00 12...VirtualBox Host-only Ethernet Adapter

15...90 65 84 94 48 65...Microsoft Wi-Fi Direct Virtual Adapter

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IPv4 Route Table

Active Routes:

Network Destination	Netmask	Gateway	Interface	Metric
0.0.0.0	0.0.0.0	172.16.96.1	172.16.102.158	30

Some important Linux networking commands

1. ip - can show address information, manipulate routing, plus display network various devices, interfaces & tunnels.

a) ip address show

1. lo: <LOOPBACK, UP, LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000

link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00

inet 127.0.0.1/8 scope host lo

valid_lft forever preferred_lft forever

inet6 ::1/128 scope host

valid_lft forever preferred_lft forever

b) ip route get ~~google.com~~ 8.8.8.8

8.8.8.8 via 10.0.2.2 dev enp0s3 src 10.0.2.15 uid 1000 cache

c) sudo ip address add 192.168.1.254/24 dev enps03

[sudo] password for kiki: root

d) ip route show

default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100 10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100

2. mtr

Matt's traceroute is a program with a command-line interface that serves as a network diagnostic & troubleshooting tool.

a) mtr googl.com

localhost.localdomain (10.0.2.15) → google.com
Keys: Help Display mode Restart statistics
Order of fields quit

Host	Packets			Pings			
~~1=gateway~~	Loss%	Snt	Last	Avg	Best	Wrst	St
1.gateway	0.0%	194	0.7	1.9	0.5	4.9	0.4
2. (waiting for reply)							

b) mtr -n google.com

localhost.localdomain (10.0.2.15) → google.com
Keys: Help Display mode Restart statistics Order of fiel
quit

Host	Packets			Pings			
	Loss%	Snt	Last	Avg	Best	Wrst	StDev
1. 10.0.2.2	0.0%	249	1.7	2.6	0.5	106.5	7.3

c) mtr -b google.com

Host	Packets			Pings			
	Loss%	Snt	Last	Avg	Best	Wrst	StDev
1. gateway(10.0.2.2)	0.0%	51	2.0	1.5	0.5	3.2	0.6

2. (waiting for reply)

3. tcpdump - it is used for capturing & displaying packets.

to install tcpdump

dnf install -y tcpdump

O/p [$ sudo] password for kcarthana: root

4. ping: best way to test connectivity between 2 nodes.

ping -c 10 google.com

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10 packets transmitted, 10 received, 0% packet loss, time 9021ms

Result: Hence various network commands were executed in Linux & Commands